

Student Name (In Arabic):

Lecture Date and Time:

Time: 2 hours/ 1st Semester; January - 2016

Instructor name:

Helpful data: $N_A = 6.02 \times 10^{23}$ species/mol; $R = 8.314 \text{ J/mol}^{\circ}\text{K}^2$; $F = 96500 \text{ C/mole } e^-$.

$$\ln \frac{k_1}{k_2} = -\frac{E_a}{R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right); \text{ where } T_1 > T_2$$

Choose the most correct answer for each of the following questions and transfer it to the table below in CAPITAL LETTERS

Question	1	2	3	4	5	6	7	8	9	10	11	12	13
Answer													

Question	14	15	16	17	18	19	20	21	22	23	24	25
Answer												

1. Consider the unbalanced redox equation:

$_ \text{H}_2\text{O} + _ \text{CoCl}_2 + _ \text{OCl}^-_{(aq)} \rightarrow _ \text{Co}(\text{OH})_3 + _ \text{Cl}^-_{(g)} + _ \text{H}^+_{(l)}$. Which of the following sets of numbers will balance the equation?

- A) 5, 2, 1, 2, 5, 4 B) 2, 1, 5, 2, 5, 4 C) 5, 1, 2, 2, 5, 4 D) 2, 1, 5, 5, 2, 4
E) 2, 1, 5, 4, 5, 2

2. Electrons are lost by the:

- A) Reducing agent as it undergoes reduction
B) Oxidizing agent as it undergoes reduction
C) Reducing agents as it undergoes oxidation
D) Oxidizing agent as it undergoes oxidation
E) Oxidizing agent as it undergoes hydrolysis

3. The oxidation number of Molybdenum (Mo) equals +2 in:

- A) MoO_4^{2-} B) Mo_2S_3 C) MoCl_5 D) $\text{Mo}_6\text{Cl}_{12}$ E) MoF_4

4. The mass (in g) of Cr (52.00 g/mol) that can be electroplated by the passage of 5.2 Amperes through a solution of $\text{Cr}_2(\text{SO}_4)_3$ for 45.0 minutes equals:

- A) 3.61 B) 2.522 C) 3.98 D) 3.007 E) 4.62

5. Consider the balanced equation: $2\text{I}_2 + \text{S}_2\text{O}_3^{2-} + 3\text{H}_2\text{O} \rightarrow 4\text{I}^-_{(aq)} + 2\text{H}_2\text{SO}_3 + 2\text{H}^+_{(aq)}$. 13.20 mL of 0.0374 M sample of $\text{S}_2\text{O}_3^{2-}$ were needed to completely titrate 10.00 mL of I^2 solution, the molarity of I^2 solution equals:

- A) 0.136 B) 0.144 C) 0.0574 D) 0.0957 E) 0.0987

6. The number of coulombs of charges that passes through a cell if 2.4 A of a current is applied for 85 minutes equals:
- A) 1.224×10^4 B) 3.78×10^3 C) 5.040×10^3 D) 8.820×10^3
 E) 7.56×10^3

7. For a first order reaction at a selected temperature, which will change?

Rate of reaction	rate constant	Concentration of reactants	Half life
I	II	III	IV
A) II, III	B) IV	C) I & III	D) II, IV
			E) I, IV

8. The reaction $2AB \rightarrow A_2 + B_2$ follows a first order kinetics, which of the following will yield a linear plot?

- A) $[AB]$ vs time B) $\ln [AB]$ vs time C) $1/[AB]$ vs time
 D) $[AB]^2$ vs time E) $\ln [\text{time}]$ vs $[AB]$

9. Which of the following statements is typically true for a catalyst?

- A) The concentration of the catalyst will go down as the reaction proceeds.
 B) The catalyst speeds up the reaction rate by decreasing rate constant.
 C) The catalyst speeds up the reaction rate by increasing the activation energy.
 D) The catalyst speeds up the reaction rate by increasing products concentration.
 E) The catalyst provides a new pathway with lower activation energy.

10. The following data were collected for the reaction $A_4 + 6B_2 \rightarrow 4AB_3$, the order of reaction with respect to A_4 equals:

- A) 1 B) 0 C) 2
 D) 3 E) 4

Run	$[A_4]_0$	$[B_2]_0$	Initial rate (M/s)
1	0.011	0.0075	3.2×10^{-4}
2	0.011	0.0150	3.2×10^{-4}
3	0.022	0.0150	6.4×10^{-4}

11. The data below were obtained for moles of A as the reaction $A_{(g)} \rightarrow \text{Products}_{(g)}$ proceeded. At a certain experiment, a flask containing 0.100 mol of $A_{(g)}$ was allowed to react, the average rate of disappearance of A between 100 and 150 seconds equals:

Time (s)	0.00	50.0	100	150	200
Moles of A	0.1000	0.0079	0.0065	0.0055	0.0048

- A) 6.67×10^{-4} B) 2.5×10^{-3} C) 2.0×10^{-5} D) 1.7×10^{-5}
 E) 2.84×10^{-3}

12. For the second order reaction $2A \rightarrow B$, the rate constant is $3.9 \times 10^{-5} \text{ M}^{-1} \cdot \text{s}^{-1}$. How long will it take (in seconds) to decrease the concentration of A from 1.07 M to 0.24 M?

- A) 260.2 B) 8.29×10^4 C) 6.303×10^4 D) 5.15×10^4
 E) 4.035×10^3

13. For the reaction $\text{BrO}_3^- + 5\text{Br}^- + 6\text{H}^+ \rightarrow 3\text{Br}_2 + 3\text{H}_2\text{O}$, at a particular time, $-\Delta[\text{BrO}_3^-]/\Delta t = 1.5 \times 10^{-2} \text{ M/s}$. What is $-\Delta[\text{Br}^-]/\Delta t$ at the same instant?

- A) 4.5×10^{-2} B) 9.0×10^{-2} C) 1.3×10^{-2} D) 5.0×10^{-3}
E) 7.5×10^{-2}

14. It takes 42.0 min for the concentration of a reactant in a first order reaction to drop from 0.45 M to 0.32 M at 25 °C. How long will it take (in min) for the reaction to be 90 % complete?

- A) 113 B) 284 C) 148 D) 369 E) 198

15. The rate constant k for the reaction $\text{A} \rightarrow \text{Products}$ equals $0.039 \text{ M}^{-1} \text{ s}^{-1}$. If the $[\text{A}]$ is 0.30 M, then the half-life (in seconds) of this reaction equals:

- A) 42.7 B) 3.85 C) 17.8 D) 85.5 E) 3.37

16. The activation energy for the reaction $\text{CH}_3\text{CO} \rightarrow \text{CH}_3 + \text{CO}$ equals 71 kJ/mol. How many times greater is the rate constant for this reaction at 170 °C than 150 °C?

- A) 1.6 B) 6.2 C) 2.5 D) 5.7 E) 1.9

17. For the following melting process $\text{A}_{(s)} \rightleftharpoons \text{A}_{(l)}$, $\Delta H_{\text{melting}} = 10.9 \text{ kJ/mol}$ and $\Delta S_{\text{melting}} = 39 \text{ J/K} \cdot \text{mol}$. The melting point (°C) for this solid equals:

- A) 6.5 B) 47.58 C) 116.3 D) 146.2 E) 272.0

18. Consider the nonspontaneous reaction $\text{PCl}_{5(g)} \rightarrow \text{PCl}_{3(g)} + \text{Cl}_{2(g)}$, if $\Delta H^\circ = 87.9 \text{ kJ/mol}$ and $\Delta S^\circ = 240 \text{ J} \cdot \text{mol}^{-1} \cdot \text{K}^{-1}$; then, this reaction will become spontaneous when the temperature is greater than ----- (°C):

- A) 215.3 B) 166.5 C) 145.57 D) 93.25 E) 126.5

19. Given the following thermochemical equations:

$2\text{Al}_{(s)} + 1.5 \text{O}_{2(g)} \rightarrow \text{Al}_2\text{O}_{3(s)}$	$\Delta H^\circ = -1669.8 \text{ kJ}$
$2\text{Fe}_{(s)} + 1.5 \text{O}_{2(g)} \rightarrow \text{Fe}_2\text{O}_{3(s)}$	$\Delta H^\circ = -822.2 \text{ kJ}$
Then, ΔH° (in kJ) for the reaction: $2\text{Al}_{(s)} + \text{Fe}_2\text{O}_{3(s)} \rightarrow 2\text{Fe}_{(s)} + \text{Al}_2\text{O}_{3(s)}$ equals:	

- A) -869.6 B) -847.6 C) -469.6 D) -689.6
E) -947.4

20. Use the data provided in the table below to find ΔH (kJ/mol) for the reaction: $\text{Br}-\text{Br} + 3\text{F}-\text{F} \rightarrow 2\text{BrF}_3$

- A) 516 B) +1032 C) -516
D) -524 E) -1032



Bond	BE (kJ/mol)
Br-Br	+190
F-F	+158
Br-F	+198

21. Which one of the following reactions has a POSITIVE entropy change?

- A) $2\text{SO}_{2(g)} + \text{O}_{2(g)} \rightarrow 2\text{SO}_{3(g)}$ B) $\text{BF}_3(g) + \text{NH}_3(g) \rightarrow \text{F}_3\text{BNH}_3(s)$
C) $\text{H}_2\text{O}_{(g)} \rightarrow \text{H}_2\text{O}_{(l)}$ D) $\text{N}_2(g) + 3\text{H}_2(g) \rightarrow 2\text{NH}_3(g)$
E) $2\text{NH}_4\text{NO}_{3(s)} \rightarrow 2\text{N}_2(g) + 4\text{H}_2\text{O}_{(g)} + \text{O}_{2(g)}$

Al-Balqa Applied University
Faculty of Engineering Technology
Department of Physics and Basic Sciences
General Chemistry (CHEM 101) Final examination

Name (in Arabic): _____
Time 120 minutes

Date: 14/5/2016
INSTRUCTOR:

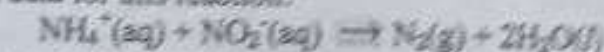
Helpful data: $A_{wt} (g \cdot mol^{-1})$ for Na = 23.0, Mg = 24.3, Al = 27.0, Fe = 55.85, C_{66}
= 40.1, O = 16.0, H = 1, N = 14, C = 12.0, $R = 8.314 J \cdot mol^{-1} K^{-1}$
 $\ln(k_2/k_1) = -E_a / R (1/T_2 - 1/T_1)$ where $T_2 > T_1$

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Answer													

Question No.	14	15	16	17	18	19	20	21	22	23	24	25	26
Answer													

Circle the most correct answer in the following questions:

1. Given the following data for this reaction:

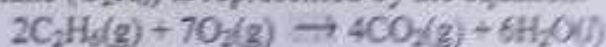


TRIAL	$[NH_4^+]$	$[NO_2^-]$	RATE
1	0.010 M	0.020 M	0.020 M/s
2	0.015 M	0.020 M	0.030 M/s
3	0.010 M	0.010 M	0.005 M/s

The rate law for the reaction is:

- (a) Rate = $k[NH_4^+][NO_2^-]$
- (b) Rate = $k[NH_4^+]^2[NO_2^-]^2$
- (c) Rate = $k[NH_4^+]^2[NO_2^-]$
- (d) Rate = $k[NH_4^+][NO_2^-]^2$
- (e) none of the above

2. The combustion of ethane (C_2H_6) is represented by the equation:

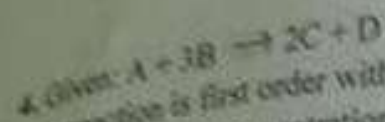


In this reaction:

- (a) ethane is consumed one half as CO_2 is formed.
- (b) the rate of formation of CO_2 equals the rate of formation of water.
- (c) water is formed at a rate equal to two-thirds the rate of formation of CO_2 .
- (d) the rate of consumption of oxygen equals the rate of consumption of water.
- (e) CO_2 is formed twice as fast as ethane is consumed.

3. What are the units of k for the rate law: Rate = $k[A]^2[B]^2$, when the concentration unit is mol/L?

- (a) s^{-1}
- (b) s
- (c) $L^3 \cdot mol^{-3} \cdot s^{-1}$
- (d) $L^2 \cdot mol^{-2} \cdot s^{-1}$
- (e) $L^2 \cdot s^2 \cdot mol^{-2}$



This reaction is first order with respect to reactant A and first order with respect to reactant B. If the concentration of A is doubled and the concentration of B is halved, the rate of the reaction would _____ by a factor of _____.

- (a) increase, 2 (b) decrease, 2 (c) increase, 4 (d) decrease, 4 (e) not change

5. If $[A]_0$ of a zero order reaction = a, then the time required for 50% conversion is:

- (a) $a/2k$ (b) ak (c) $2k/a$ (d) k/a (e) a/k

6. The original concentration of a first-order reaction is 0.84 M. What is the half-life if, after 2.0 minutes, the reactant concentration is 0.062 M?

- (a) 4 s (b) 64 s (c) 8 s (d) 32 s (e) 16 s

7. If the activation energy in the forward direction of a reaction is 74 kJ and the activation energy in the reverse direction is 52 kJ, what is the ΔH value for this reaction?

- (a) 22 kJ (b) -22 kJ (c) 52 kJ (d) -52 kJ (e) 126 kJ

8. By which factor the rate of a reaction increased upon raising the temperature from 300 K to 360 K, if the activation energy of the reaction is 69 kJ.

- (a) 27 (b) 35 (c) 100 (d) 53 (e) 50

9. Which of the following is correct about the catalyst?

- (a) it increases the energy barrier between reactants and products.
(b) it changes the equilibrium concentration of the products.
(c) it does not affect a reaction energy path.
(d) it always decreases the rate for a reaction.
(e) None of the above

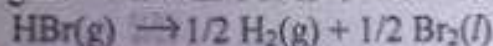
10. Which of the following thermodynamic quantities is a state function?

- (a) Gibbs free energy (b) enthalpy (c) entropy (d) internal energy (e) All

11. At a constant temperature, an ideal gas is expanded from 4.0 liters to 6.0 liters against a constant external pressure of 5.0 atm. How much work is done?

- (a) $w = +10$ liter atm (b) $w = -10$ liter atm (c) $w = +30$ liter atm
(d) $w = -30$ liter atm (e) The answer cannot be calculated.

12. The ΔH° for the following reaction at 298 K is -36.4 kJ.



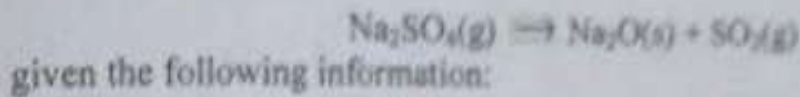
Calculate ΔE° at 298 K. The universal gas constant, R, is 8.314 J/mol K.

- (a) -35.2 kJ (b) +35.2 kJ (c) -36.4 kJ (d) -37.6 kJ (e) +37.6 kJ

13. For which of the following reactions would the ΔH° for the reaction be labeled ΔH_f° ?

- (a) $\text{PCl}_3(\text{g}) + 1/2 \text{O}_2(\text{g}) \rightarrow \text{POCl}_3(\text{g})$
- (b) $1/2 \text{N}_2\text{O}(\text{g}) + 1/4 \text{O}_2(\text{g}) \rightarrow \text{NO}(\text{g})$
- (c) $\text{CaO}(\text{s}) + \text{SO}_2(\text{g}) \rightarrow \text{CaSO}_3(\text{s})$
- (d) $\text{Al}(\text{s}) + 3/2 \text{H}_2(\text{g}) + 3/2 \text{O}_2(\text{g}) \rightarrow \text{Al}(\text{OH})_3(\text{s})$
- (e) The ΔH° for all these reactions would be labeled ΔH_f° .

14. Calculate ΔH° for the reaction:



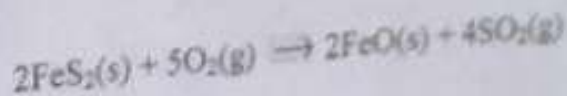
- | | ΔH° |
|--|------------------|
| (1) $\text{Na}(\text{s}) + \text{H}_2\text{O}(\text{l}) \rightarrow \text{NaOH}(\text{s}) + 1/2 \text{H}_2(\text{g})$ | -146 kJ |
| (2) $\text{Na}_2\text{SO}_4(\text{s}) + \text{H}_2\text{O}(\text{l}) \rightarrow 2\text{NaOH}(\text{s}) + \text{SO}_2(\text{g})$ | +418 kJ |
| (3) $2\text{Na}_2\text{O}(\text{s}) + 2\text{H}_2(\text{g}) \rightarrow 4\text{Na}(\text{s}) + 2\text{H}_2\text{O}(\text{l})$ | +259 kJ |

- (a) +255 kJ (b) -435 kJ (c) -581 kJ (d) +581 kJ (e) -452 kJ

15. In the ground state of a cobalt $_{27}\text{Ni}$ atom there are _____ unpaired electrons and the atom is _____.

- (a) 3, paramagnetic
- (b) 5, paramagnetic
- (c) 2, diamagnetic
- (d) 0, diamagnetic
- (e) 2, paramagnetic

16. Calculate ΔH_{rxn} for the reaction:



given the following information:

$$\Delta H_f^\circ \text{ for } \text{SO}_2(\text{g}) = -297 \text{ kJ/mol}$$

$$\Delta H_f^\circ \text{ for } \text{FeO}(\text{s}) = -268 \text{ kJ/mol}$$

$$\Delta H_f^\circ \text{ for } \text{FeS}_2(\text{s}) = -177 \text{ kJ/mol}$$

- (a) -1550 kJ (b) -1370 kJ (c) -774 kJ (d) -686 kJ (e) +808 kJ

17. Estimate the boiling point of $\text{Br}_2(\text{l})$ ($\Delta H = 30.9 \text{ kJ}$; $\Delta S = 93.0 \text{ J/K}$).

- (a) 59°C
- (b) 85°C
- (c) 373°C
- (d) 177°C
- (e) 44°C

18. For the reaction, $\text{A} + \text{B} \rightarrow \text{C}$, $\Delta H^\circ = +30 \text{ kJ}$; $\Delta S^\circ = +50 \text{ J/K}$. Therefore the reaction is:

- (a) spontaneous at temperatures greater than 600 K.
- (b) spontaneous at all temperatures.
- (c) non-spontaneous at all temperatures.

- (d) spontaneous at temperatures less than 600 K
 (e) spontaneous only at 25°C.

19. What is the charge on the copper (Cu) ion in, $\text{Cu}_3(\text{CO}_3)(\text{OH})$?
 (Hint: MgCO_3 and $\text{Mg}(\text{OH})_2$ are common salts)

- (a) 2+ (b) 1+ (c) 0 (d) 1- (e) 2-

20. Balance the following equation using minimum integral coefficients:



The stoichiometric coefficient for H_2O is:

- (a) 6 (b) 4 (c) 3 (d) 7 (e) 5

21. How many milliliters of 0.200 M NH_4OH are needed to react with 24.0 mL of 0.550 M FeCl_3 ?



- (a) 99.0 mL (b) 33.0 mL (c) 8.25 mL (d) 68.8 mL (e) 198 mL

22. How many mL of 17 M NH_3 must be diluted to 250.0 mL to make a 0.75 M solution?

- (a) 11 mL (b) 22 mL (c) 39 mL (d) 73 mL (e) none of these

23. Which element has the **highest** first ionization energy?

- (a) ${}_2\text{He}$ (b) ${}_{10}\text{Ne}$ (c) ${}_{18}\text{Ar}$ (d) ${}_{36}\text{Kr}$ (e) ${}_{54}\text{Xe}$

24. All of the following properties of the alkaline earth metals increase going down the group **except**

- (a) atomic radius (b) atomic volume (c) ionic radius
 (d) atomic mass (e) none of the above

25. Which of these isoelectronic species has the **largest** radius?

- (a) ${}_{35}\text{Br}^-$ (b) ${}_{38}\text{Sr}^{2+}$ (c) ${}_{37}\text{Rb}^+$ (d) ${}_{34}\text{Se}^{2-}$ (e) All have the same size

26. A 1.0 g sample of a compound containing only antimony Sb (Awt = 122 g/mol) and oxygen was found to contain 0.836 g of antimony and 0.164 g of oxygen. What is the simplest formula for the compound?

- (a) SbO (b) SbO_2 (c) Sb_2O_3 (d) Sb_3O_4 (e) Sb_2O_5